



Marwadi
University
Marwadi Chandarana Group



Branch/Semester	B.Tech CSE (AI & ML) / Semester V
Subject Name:	Design and Analysis of Algorithms
Subject Code:	01AI0506
Assignment:	Assignment No. II (Result Improvement Programme)
Date:	9 th September 2026
Faculty Name:	Prof. Abhishek Chauhan

Sr.	Sorting & Searching Techniques and Dynamic Programming	CO	BL
1	Apply Bubble Sort to arrange a given set of elements in ascending order. Show the array after each pass.	CO2	BL3
2	Apply Insertion Sort to arrange a given set of elements in ascending order. Show the result after each insertion.	CO2	BL3
3	Apply Quick Sort to sort a given set of transaction values. Show the major partitioning steps and the final sorted array.	CO2	BL3
4	Apply Merge Sort to a given array. Draw the division structure and show how the sorted subarrays are merged.	CO2	BL3
5	Apply Binary Search to locate a given element in a sorted array. Show the values of low, high, and mid at each step.	CO2	BL3
6	Analyze the best-case, worst-case, and average-case time complexities of the Linear Search algorithm for successful and unsuccessful searches.	CO3	BL4
7	Apply the Making Change algorithm using Dynamic Programming for a given denomination set and amount. Construct the complete DP table and identify the coins used.	CO4	BL3
8	Apply the 0/1 Knapsack Dynamic Programming algorithm for given items and capacity. Construct the complete DP table and determine the maximum profit along with selected items.	CO4	BL3
9	Apply the LCS Dynamic Programming algorithm to find the Longest Common Subsequence of two given strings. Show the complete LCS table and state the LCS.	CO4	BL3

10	Judge whether Dynamic Programming is an appropriate technique for solving the LCS and Matrix Chain Multiplication problems. Defend your answer by analyzing optimal substructure and overlapping subproblems, and compare time complexity with a naive recursive approach.	CO4	BL5
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